Jongseok Kim

kjseok@etri.re.kr Scholar Portfolio +82-10-3948-5699

RESEARCH INTERESTS

- Machine learning for heterogeneous and multimodal data, with a focus on representation learning, efficient models, and generative modeling.
- **Keywords:** Multimodal Learning, Representation Learning, Efficient AI, Generative Models, LLMs

EDUCATION

M.S. in Computer Science

Chungbuk National University 2023/03 – 2025/02

Cheongju, South Korea

- Advisor: Prof. Ohyun Jo
- Thesis: *Learning Framework for Enhancing Complex-Valued Sequence Data Processing via Multi-Shape Augmentation*
- **Outstanding Graduate Researcher Award** (awarded annually to one graduate student in the department)

B.S. in Computer Science

Chungbuk National University 2017/03 – 2023/02

Cheongju, South Korea

- Including mandatory military service, Republic of Korea Army (2018/07 – 2020/03)

PUBLICATIONS

* Corresponding author † Co-first authors

International Journals and Conferences

- Restoration of Noisy Reference Signal by Using Diffusion Model
  Jonghwan Shin, **Jongseok Kim**, Ohyun Jo*
  IEEE Wireless Communications Letters, 2026

- ComplexRep: Integrating Learned Representations to Enhance Complex-valued Data Transparency
  Jongseok Kim, Woonggyu Min, Juyeop Kim, Ohyun Jo*
  IEEE Internet of Things Journal, 2026

- Denoising Method for Wireless Communication Signals Based on Convolutional AutoEncoder
  Woonggyu Min, **Jongseok Kim**, Ohyun Jo*
  ICAIIC 2025

- MuShAug: Boosting Sequence Signal Classification via Multishape Augmentation
  Jongseok Kim, Ohyun Jo*
  IEEE Internet of Things Journal, 2024

- IncepSeqNet: Advancing Signal Classification with Multi-Shape Augmentations (Student Abstract)
  Jongseok Kim, Ohyun Jo*
  AAAI 2024

- Intelligent Index Classification Method Based on Machine Learning for Detection of Reference Signal in 5G Networks
  Seungwoo Kang†, Taegyeom Lee†, **Jongseok Kim**, A-reum-saem Lee, Juyeop Kim, Ohyun Jo*
  IEEE Access, 2023

Preprints and Under Review

- LeMoF: Level-guided Multimodal Fusion for Heterogeneous Clinical Data
  Jongseok Kim†, Seongae Kang†, Jonghwan Shin, Yuhan Lee, Ohyun Jo*
  arXiv:2601.10092, 2026

- Hybrid Feature Selection for Assessment of Oceanic Channel via Explainable AI
  Jongseok Kim, Ho-Shin Cho, Ohyun Jo*
  Journal of Ocean Engineering and Science, Major Revision

- Title withheld (double-blind review)
  Jongseok Kim[†], Byunghyuk Youn[†], Ohyun Jo*
  IEEE Transactions on Mobile Computing, Under Review

- Title withheld (double-blind review)
  Jongseok Kim, Seongae Kang, Ohyun Jo*
  ICLR 2027, Under Review

- Are All Augmented Views Equally Useful? RD-Bank for Complex-Valued Data Augmentation
  Jongseok Kim, Byunghyuk Youn, Woonggyu Min, Suwoong Lee, Ohyun Jo*
  ICASSP 2027, Under Review

- From Processed to Raw: Importance-Guided Feature Distillation across Audio Representations
  Jongseok Kim, Jonghwan Shin, Suwoong Lee, Ohyun Jo*
  ICASSP 2027, Under Review

- Title withheld (double-blind review)
  Seongae Kang, **Jongseok Kim**, Jee Hyun Kim, Yuhan Lee, Ohyun Jo*
  ACM SAC 2027, Under Review

Additionally, 12 papers in Korean domestic journals and conferences (including 2 Best Paper Awards); full list available upon request.

RESEARCH AND WORK EXPERIENCE

Post-Master's Researcher
ETRI **2026/08 – Present**
Daejeon, South Korea

- Spatial Content Research Division, Electronics and Telecommunications Research Institute
- Developing multimodal-to-CAD generation models that translate image and text inputs into executable CAD programs (CadQuery).
- Investigating large language models for structured, parametric CAD program generation.

Research Assistant
Chungbuk National University **2025/03 – 2026/08**
Cheongju, South Korea

- Information System Lab (Advisor: Prof. Ohyun Jo)
- Led research on representation learning and data augmentation for complex-valued signals (ComplexRep, RD-Bank).
- Developed multimodal fusion for heterogeneous clinical data (LeMoF) and cross-representation knowledge distillation for audio.

Software Developer
Planit Co., Ltd. **2021/03 – 2021/12**
Cheongju, South Korea

- Developed medical imaging software.

RESEARCH PROJECTS

Development of a Multimodal Recognition-based AI Authoring Platform Technology for Creating Installation Art Structures
Ministry of Culture, Sports and Tourism **2026/08 – Present**
South Korea

- Multimodal CAD generation and restoration

Integrated Underwater Surveillance Research Center for Future Technology Adaptation
Ministry of National Defense **2024/09 – 2026/08**
South Korea

- Explainable AI for underwater channel assessment

Ultra-Lightweight Machine Learning based on 3D Imagification of Heterogeneous Time-Series Data for IoIT Convergence Services

National Research Foundation of Korea 2022/03 – 2026/08

South Korea

- Time-series imagification and lightweight models

Development of 5G+ Intelligent Base Station Software Modem

Ministry of Science and ICT 2022/03 – 2025/12

South Korea

- Lightweight deep learning for 5G signal detection

AWARDS AND HONORS

Best Paper Award

KIIS Spring Conference 2026

South Korea

Best Paper Award

KICS Winter Conference 2026

South Korea

Outstanding Graduate Researcher Award

Chungbuk National University 2025

South Korea

- Awarded annually to one graduate student in the Department of Computer Science

PATENTS

- Method and Apparatus for Constructing a Training Dataset Comprising Text Annotations and Parametric CAD Code, Grounded in Target-Domain Shape Samples
  Invention disclosure submitted to ETRI, 2026 (patent application in preparation)
- Method for Augmenting Time-Series Signal Data for Deep Learning and Computing Device for Executing the Method
  Korean Patent Application No. 10-2024-0071748, filed May 2024

TEACHING EXPERIENCE

Teaching Assistant

Chungbuk National University 2023 – 2024

Cheongju, South Korea

- Operating Systems (Spring 2023)
- Computer Networks (Spring 2024)